

Article D7: Project Management Year 5

PM.1.1 Project Management Objective

The topic of Program Management (PM) goes by many names in various industries: Project Engineering (PE), Project Management (PM), Product Management (PM), Program Management (PGM), etc. To familiarize you with these terms, we will use them interchangeably.

The goal of the Project Management challenge is to have the teams use tools to better manage and execute their plan for the AutoDrive challenge.

For Year 5, the structure of this challenge has evolved and changed. **The five core principles from previous years have been combined to become 4 core principles (Team Structure and Team Communication have been combined). Also, the Winter Workshop format has shifted from a written report to a live demonstration / presentation of these principles with Q&A.**

The Winter Workshop will now require teams to present their four project management tools—Gantt Chart, Work Breakdown Structure, Team Structure and Communications, and the Decision-Making Model, along with a Summary — in a live presentation format. It is expected that the PM will pull up a live view of the tool being discussed (i.e. during the Gantt discussion, the live Gantt chart and Critical Path should be pulled up and displayed, same for WBS). Judges will evaluate both the content and the team's ability to clearly communicate their approach and exemplified proficiency of each tool. Each section will have a 5 min presentation with a 5 min Q&A about that section along with a summary 5 min presentation and an extended Q&A after that. This will be scheduled to be 1 hour long.

The level of detail and rigor expected in each tool remains unchanged from previous years. Teams should not assume that the presentation format makes the deliverables easier; in fact, expectations for clarity, completeness, proficiency, and professionalism are higher.

Analyze and summarize the following for EACH deliverable in your winter workshop **presentation**:

- Show the deliverable live – i.e. pull up your real Gantt chart, Decision Making Model, etc.
 - You could use snips in a PowerPoint for the presentation to highlight a certain event but the live version must be up and active so it can be referenced for the judges during Q&A
- Compare what you did last year to what you have for this year. What continued, what was deprecated, what is new? Why did you keep it or why is it different? Basically, what did you change and why?
- Evaluate the tools you used. Are they the same as last year? Why / why not?
- What went well? What didn't go well?
- Based on last year's performance, what did you change in PM to address any & all shortcomings?

Format: 1 hour Time Slot: You will walk through each deliverable pausing between each for judges questions. You can use up to 5 minutes to present your deliverable before the judges stop you for questions. You do not need to use the full 5 minutes for each. However, some of these sections could be challenging to discuss in detail in 5 minutes. So, practice and focus on the rubric deliverables.

- Intro- up to 5 minutes
- Gantt Chart -up to 5 minutes
 - Judges questions – 5 minutes
- WBS- up to 5 minutes
 - Judges questions 5 minutes

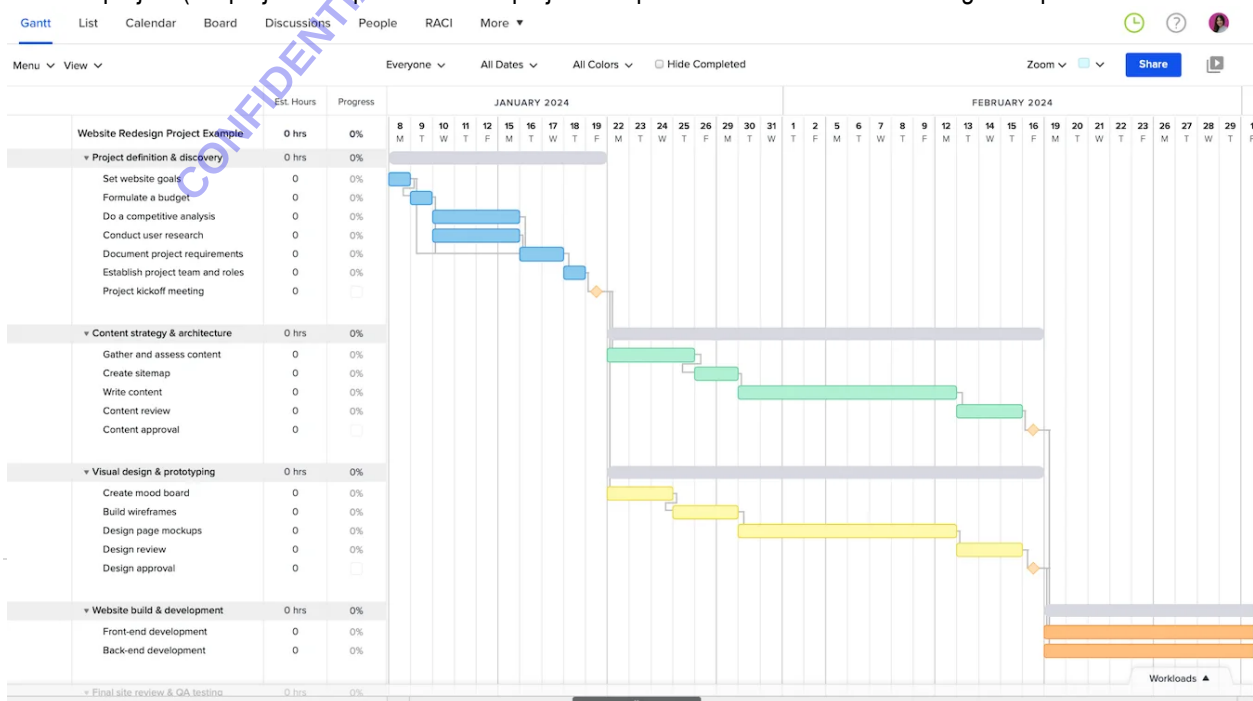
-Continued for other deliverables.....
- Summary – up to 5 minutes
 - Final comments/questions from the judges

At the Final Competition, teams will submit a written report (same format as Year 4) with an added feedback section, and deliver a presentation that reflects back on their project management journey / team journey over the past 5-years.

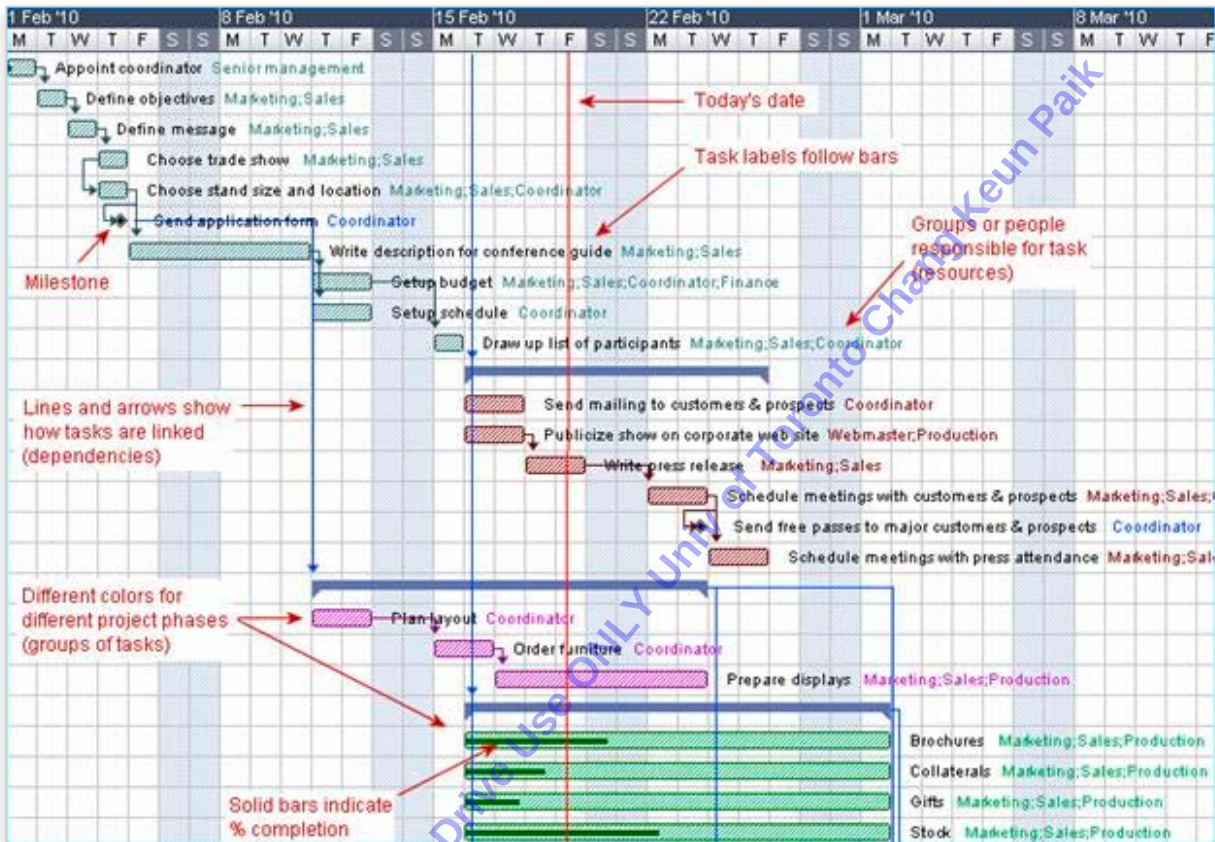
PM.1.2 Project Management Deliverables

PM.1.2.1 Gantt Chart

The first goal of a successful Project Engineer is to understand the scope and timing of the project. The easiest way to view this is with a Gantt chart, which is sometimes called a bar graph timing chart. This is a very simple concept where the deliverables are considered single date milestones (typically represented with a diamond symbol) and the bar graphs represent the time expected to complete the work associated with that milestone. Gantt charts and Work Breakdown Structures (WBS) work hand in hand and are sometimes interchangeable in meaning and descriptions. WBS will be covered in the next section. Some of the milestones will be obvious from the other parts of the AutoDrive Challenge's deliverables (all AutoDrive Deliverables, workshops, and challenges should be Gantt Chart milestones), and some milestones will be up to the individual teams to mark interim goals or review points. Usually, each milestone will require multiple different work streams (bar graphs) to be completed to successfully achieve the milestone. These can be sequential efforts, and these can be parallel efforts. Most often having a combination of parallel and sequential activities required to complete a single milestone. Understanding these relationships in a visual representation is the goal of the Gantt chart. Typically, the milestones will be represented as a diamond shape and the work element bar graphs will be connected as they cascade via arrows to indicate required predecessors and successors or what is commonly called relationships. Connectivity of all tasks is imperative and must be 100% complete; every task must be linked and cascaded down to a milestone. [Even tasks that appear to be a dead-end task (i.e. turn in report), must cascade down to the next or final task of the project. The logic here is that the project would not be complete if that didn't happen.] Then milestones can be cascaded and linked to other tasks or milestones. Basically, every single item in the Gantt chart should be linked to following items, with the only exception being the final milestone of the project (i.e. project competition and/or project complete. Please see the following examples.



Gantt Chart Example 1

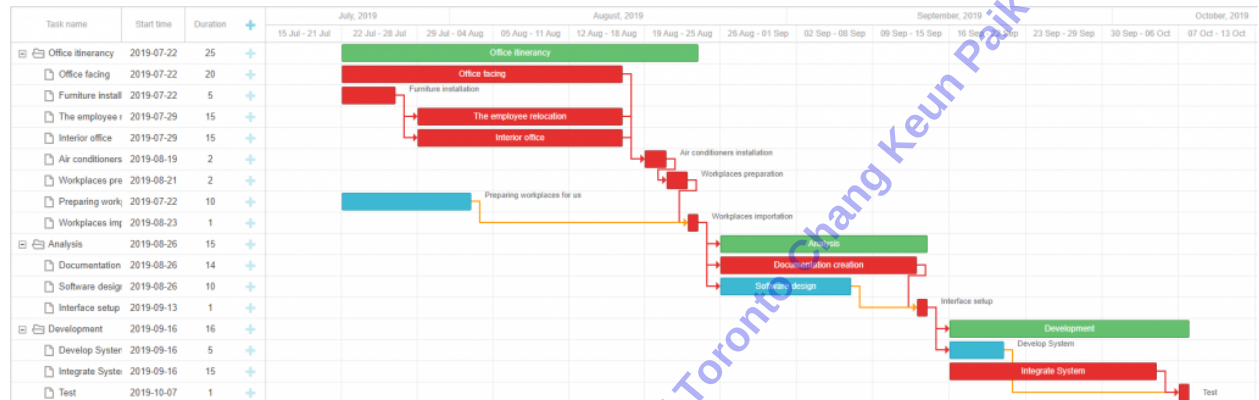


Gantt Chart example 2

These 2 examples are very simple projects just intended to give a visual reference. Your actual project timeline should be significantly larger and cover 5 epochs, or what will be the 5-year challenge. It is expected that each year will have its own visualization of a Gantt chart (a stand-alone Gantt chart), but these can be linked together to show a 5 year-long project that achieves the final goals. (Of course, this may not be a continuous Gantt chart if the Gantt chart tool has changed significantly over the 5 years; changing tools will create a natural break in the 5-year project Gantt.) It will be expected that at least year 5 is very detailed, laying out each individual workstream. All long-term projects work this way. As you move forward in time, more information is learned, and the Gantt chart gets updated to reflect the current level of information. Therefore, Gantt charts should be living documents that grow and get modified constantly. The key to good Project Engineering is accurately tracking these changes and understanding why these changes are occurring.

One of the most important parts of a well-constructed Gantt chart is highlighting and understanding the "Critical Path". The critical path is the most "at risk" chain of events that are driving key (critical) project

milestones. This is typically the chain of tasks that is the tightest packed that are critical to the key deliverables. Highlighting and tracking the critical path is very important to good project engineering. Critical paths are not the entire project! The Critical path should be a small section of the entire Gantt. Critical paths are the tasks that, if they slip at all, then the project is delayed or late. The Critical Path should also be shown on the Gantt chart and is typically shown as red, to indicate that this is the Critical Path. See Fig 3 below.



Gantt Chart Example 3 (Critical Path)

Also note that the Critical Path can span across sections of the Gantt Chart, it is possible that one section will be a requirement or predecessor of another section, and those two interactions could be on the Critical Path. A Critical Path analysis is one of the key elements or analysis of a good Project Engineer.

A good Critical Path analysis will include: What are the key tasks that are driving the project timeline? Are these key tasks on time, or are they late? If they are late, why? Also, if they are late, what is the team doing to get back on track? What is the team doing to ensure that no further slips occur? What could have been done differently to prevent this in the future? How has this impacted the Work Breakdown allocation of time? Has the Critical Path stayed the same, or has other elements now joined or become part of the Critical Path, remember a Gantt Chart is a living artifact that is constantly being updated! It is important that not only is the Critical Path identified but is discussed and analyzed.

Year-5 will be different than the previous years. This year, the Gantt chart will be evaluated in depth only once during the Winter workshop. The Year End Challenge will just review any changes that occurred. Even though there will be no written document for the Winter Workshop, it is expected that the teams will still maintain the Gantt chart and Critical Path and are constantly monitoring these and making adjustments to reflect reality. The Year End deliverable will be discussed in a later section.

Note: There are many Gantt chart tools (WBS tools) available online that work significantly better than just doing this in Excel. It is highly recommended that each team acquires a Gantt chart tool that also ties the WBS elements and Gantt chart together.

PM.1.2.2 Work Breakdown Structure

The WBS or Work Breakdown Structure is part of the Gantt chart from the previous section. A project timeline Gantt chart without appropriate WBS assignments is meaningless. A simplistic view of these two

would state that: 1. The Gantt chart is a list of when things need to be done (milestones) and what work is required to meet these milestones (this is where the WBS and Gantt charts overlap). 2. The WBS is a list of what will be or is being done and, more importantly, what resources are required to do that work. For most WBS elements this will be the people doing the work and tracking their time. The WBS is a tool meant to help plan and optimize resource utilization. If used effectively, it should be incorporated in planning what resources will be used for a task prior to beginning work. The WBS should also be regularly monitored during task execution to ensure that resources are appropriately utilized. For example, if person #1 can only work on a project for 8 hours a day but only 2 days a week and person #2 can work on a project 2.5 hours per day but just 3 days a week, person #1 can produce 2 times the work as person #2 (assuming the skill set and work level is the same). This can also be helpful in balancing workload for members of the team. The WBS can help identify if a specific person is being over or underutilized. This is not meant to be a timecard but as a resource that the project management team can leverage to assist in forecasting and balancing workload. The WBS combined with the Gantt can assist with planning work needed from members of the team and how many people will be required to reach a deliverable. This all gets reflected in the WBS.

There are many different ways of tracking WBS elements. AutoDrive will not force a particular method. These are dictated usually by the software being used.

The key points will be:

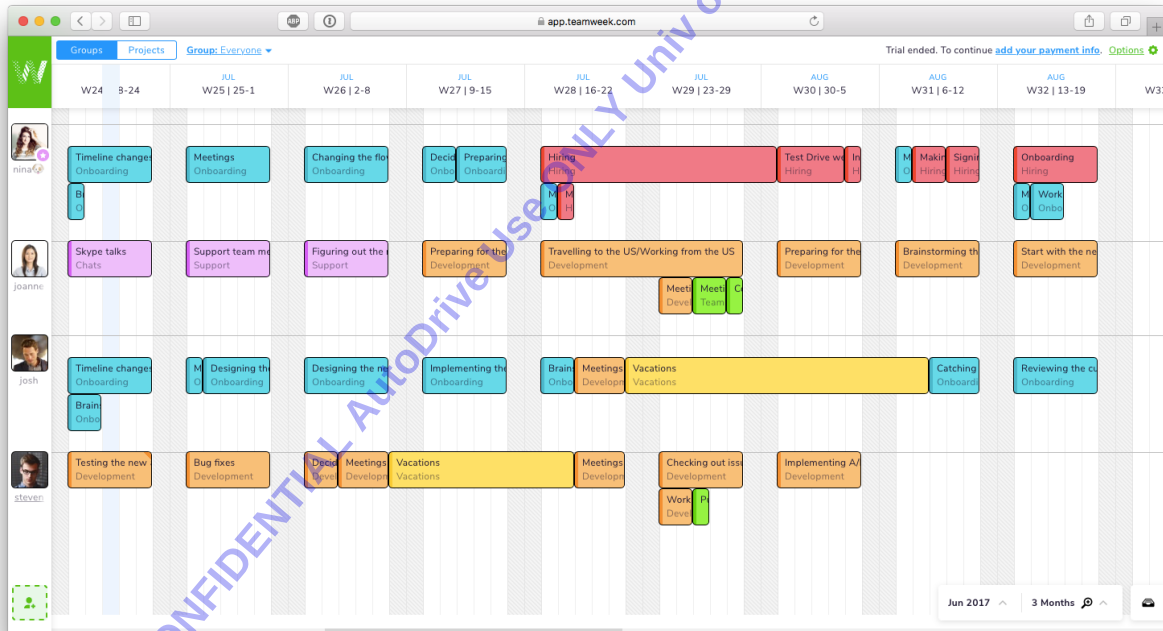
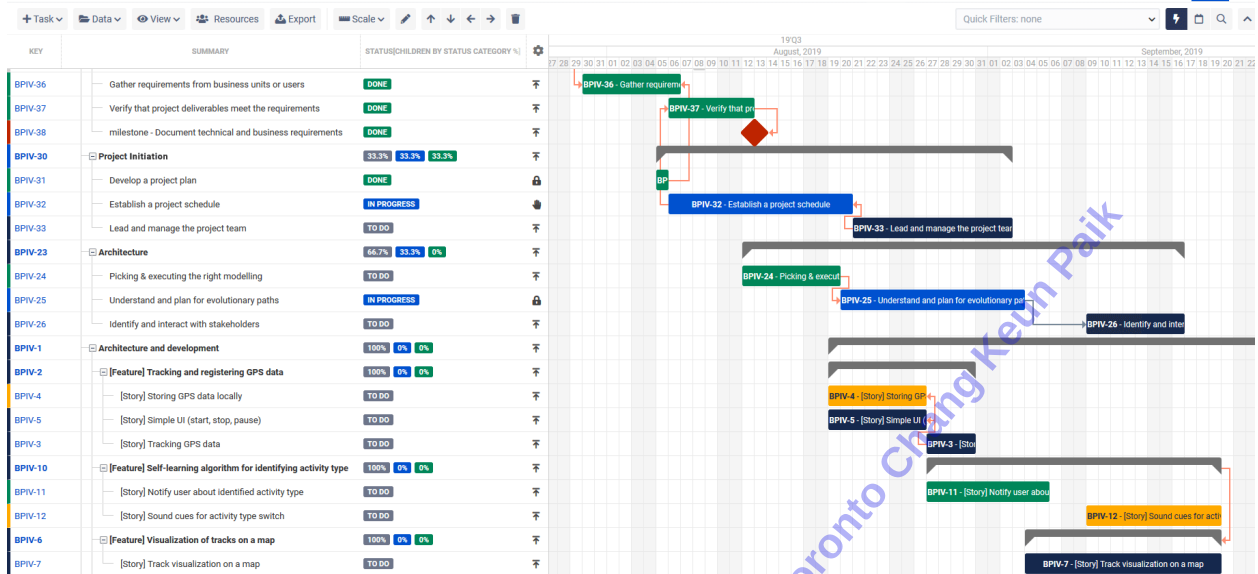
- Comprised of a comprehensive deliverable list
- Has individual names/group names assigned to each work package, if group names are used, then those groups must list all team members (required for last bullet).
- Includes hrs/wk per person
- Is tied to the Gantt Chart bar line items (line by line)
- Needs to be formatted in a way that it's easy to visualize and balance individual's workloads. Includes an overview of each team member's workload, shown week by week in hours/week
- Evidence the WBS was utilized during deliverable planning

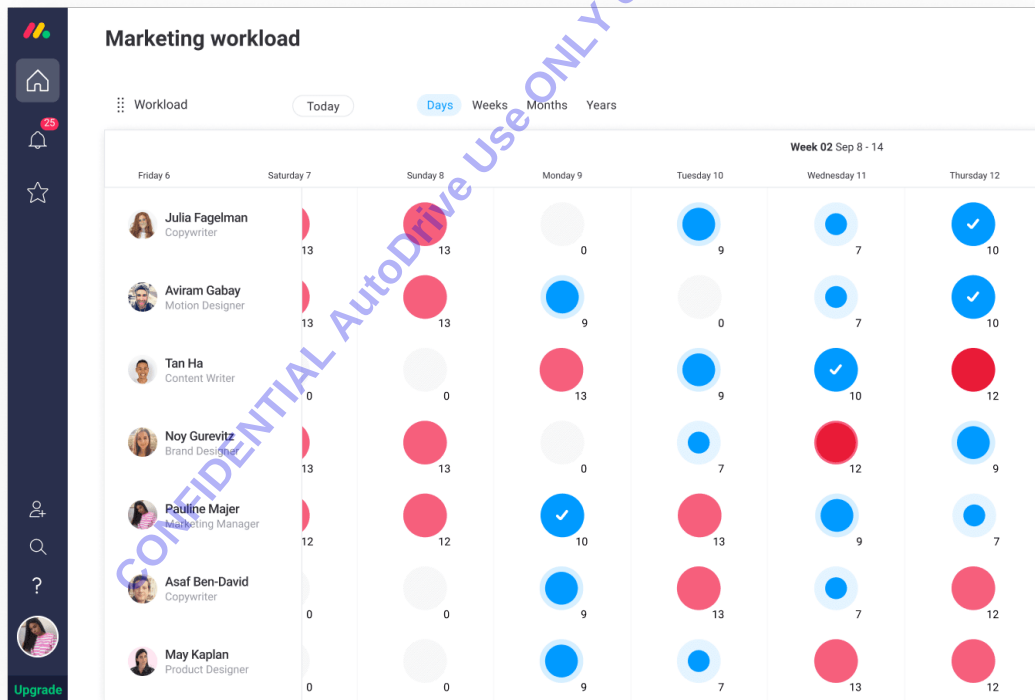
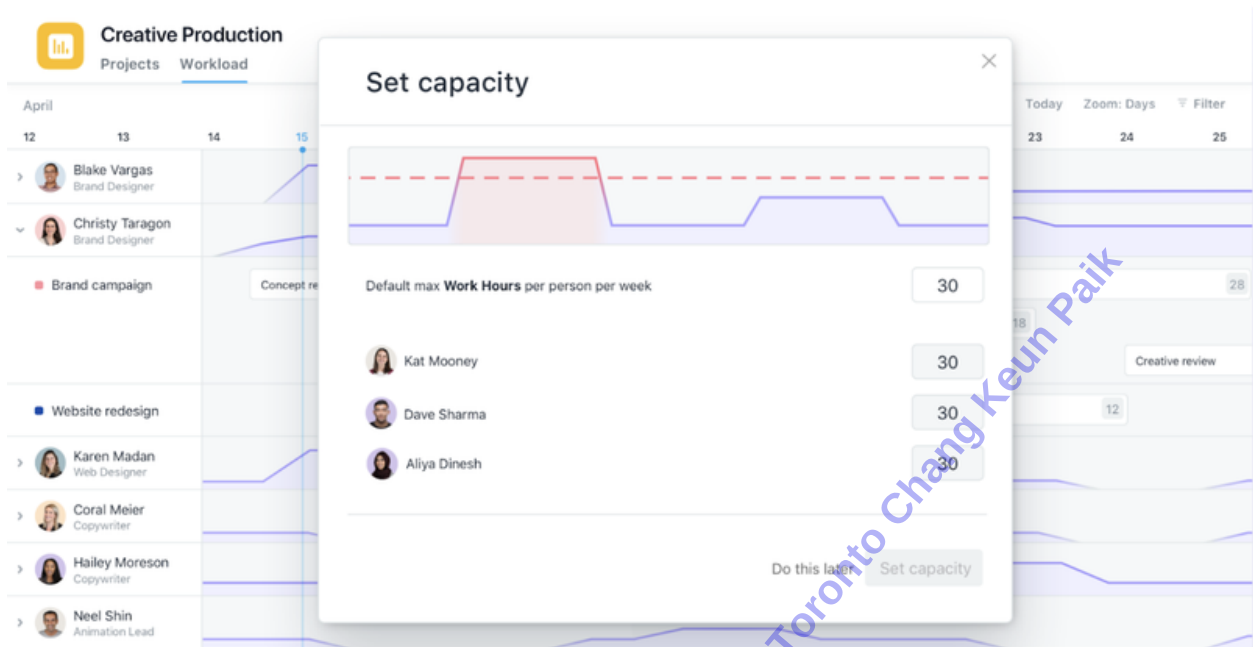
Like the Gantt chart, the WBS is a living document, constantly tracking changes. The WBS documentation will be required at each of the previously mentioned Gantt chart deliverables.

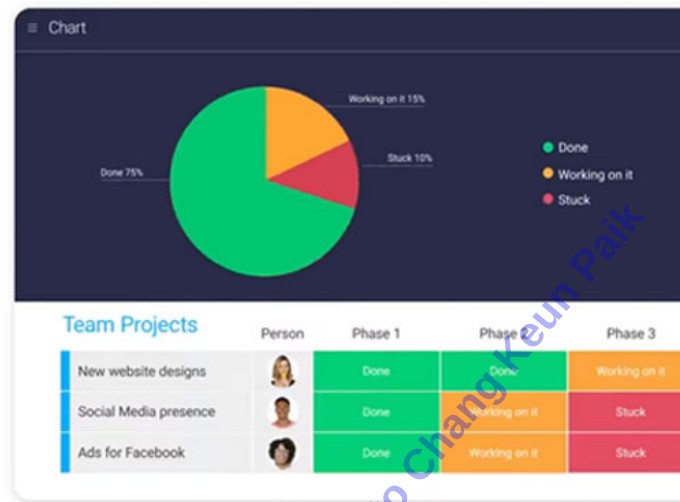
As stated before, the WBS deliverable for Year-5 will be different than previous years. The entire WBS will only be evaluated once during the Winter workshop. Even though there will be no written document for the Winter Workshop, it is expected that the teams are utilizing the WBS for planning and will consistently monitor it during deliverable execution to make adjustments. The judges will be looking for evidence of both aspects. The Year End deliverable will be discussed in a later section.

Following are examples of WBS.

The most popular Gantt chart for Jira







PM.1.2.3 Team Structure & Communication Plan

Having a team structure is important for determining who is ultimately responsible for different areas of the project and can help newer students on the team figure out how the team is run. Having a communication plan is important for keeping everyone on the team informed on what is happening within the team and for ensuring important data is not lost. For this section you will need to define your team's roles, structure, and communication plan.

All the possible roles on your team should be listed along with the responsibilities of each of those roles. Indicate how a person can attain each role such as being elected to the role, being given the role by a leader, earning the role through completing tasks, being recruited to the role for having a particular skillset, etc. Also include what (if any) requirements exist for gaining and maintaining a role and how long each role will last.

An organizational chart should be created that shows how the defined roles interact with each other and who fills those roles. It should also show how sub teams are coordinated with the team as a whole and who the team's project manager and other team leaders are. Ultimately, this chart should include every person that is on the team. In some cases, multiple people can have the same role (i.e., everyone on a sub team might have the same sub team role) or one person can have multiple roles (i.e., the Project Manager might also be the Chief Pizza Officer). Care should be taken to make sure that one person does not have too many roles. Please indicate what degree and grade level of education each member of your team is currently working towards and number of years they have been involved in AutoDrive (including if student participated in AutoDrive series 1). For example, John Smith Sophomore B.S. Electrical Engineering, AutoDrive series 2 years 2-3.

There are many different types of organizational structures that are commonly used in business such as line staff, matrix, projectized, etc. You are free to choose whichever organizational structure you think will fit best with your team and what roles you will have on your team (there is no set template). However, you are required to always have a "project manager" role whose responsibilities include at a minimum the project management deliverables. If your project manager leaves the team an interim project manager must immediately be named until a permanent replacement can be found. Supporting individuals that are not directly on the team such as faculty advisors, industry advisors, lab technicians, and others can also be included in the roles and organizational chart, however, as it is a

student run project their responsibilities should be limited to advising the team only and not include decision making, completion of tasks, etc.

Create a communication plan for your team that answers the basic questions of how you communicate within the team that can be referenced by your team members if they are unsure where/who to share information with or where to find information. Your plan should include informal, formal, documented, and external communications.

For informal communications describe how day-to-day communication is conducted between individual members of your team or smaller groups on your team. Any platforms used by the team should be listed (i.e., messaging apps, team chat software, texting, email, snail mail, social media, Kanban board, etc.). If multiple platforms are used describe how you choose what is communicated on each platform. Your formal communications can include communications that are sent to the full team, any official team meetings, and team schedules. List out what platforms are used for your team's formal communications (these could be the same as informal) and what information is planned to be communicated on those platforms. Also create a schedule that shows reoccurring meetings and any designated work times which could include open lab time, test track time, sub-team work time.

Documented communications are any information or knowledge that needs to be stored long term. This can include software / code, simulation models, CAD models, submitted documents, lessons learned, test data, etc. Describe your change control process and document retention plan for any information or data that needs to be stored. Include information on what tools are used (Git, SVN, network drives, etc.), who has access to those tools, and what (if any) review process is done to update those files. Provide examples/proof of your change control process being used throughout the year.

External communications are any communications between the team and persons outside of the team. These can include recruitment efforts, sponsor engagement, and public communications. Describe in detail how you recruited new students to be on your team. This could include efforts to recruit students with specific talent or more general recruitment efforts. For sponsor engagement describe how you communicated with your existing sponsors and what (if any) efforts were made to recruit new sponsors. List out any public social media channels that your team has and describe what each is used for. If your team is involved in any other public communications such as attending community events, television interviews, or something similar describe what was done and how it benefited the team or your community.

PM.1.2.4 Decision-Making Model (DMM)

A decision-making model is an important tool for teams to use because it shows how information should flow and where decisions get made. This model helps teams define who needs to be involved when making a decision, and what decision can be made individually/without the entire team. The key to this is finding a process that doesn't slow your progress down because every decision needs to be vetted through a large number of people, but also making sure that decisions that affect other parts of the system are being communicated.

Your decision-making model should tie into the team structure, showing each sub-team. Team members should be able to reference this chart at any point to help them understand: (1) who needs to be involved in the decision-making process for a specific item and (2) who gets the final say in the decision. Often, in the corporate world, a DMM differs from the organizational chart. However, in this case, it's possible that your organizational chart is the same as the DMM. Decisions are not only technical decisions but also team management decisions- i.e. who should fill a specific role or do a specific task?

Along with creating DMM, your team should be able to explain it and how the team uses it. This should include who has the final say in all disputes and who reconciles sub team debates. If your DMM is the same as your organizational chart, indicate the levels of decisions. Which subgroups can make which decisions and how do these subgroup decisions get aggregated into the overall project? If your DMM is different from your organizational chart, please explain and elaborate on any differences.

You must show a visual representation of the complete decision-making model. The team should include examples of how the model was used throughout the year.

PM.1.2.6 Summary

This section applies only to the Final report and presentation. The summary should provide a high-level overview of the teams' year 5 performance using PM terms. Describe in detail the health of the team and how you feel the team is doing. Explain what went well and did not go well for year 5. Additionally, the team should include how they did over the five years of series 2. Was the team satisfied with their performance, why or why not? Looking back, what would the team have done differently if you could?

PM.1.3 Project Leadership Format and Judging

PM.1.3.1 Scoring Process

Winter Workshop

Your project manager (or the interim) must be in attendance at the winter workshop and be the main presenter.

The Winter Workshop will now require teams to present their four project management tools—Gantt Chart, Work Breakdown Structure, Team Structure and Communications, and the Decision-Making Model, along with a Summary — in a live presentation format. It is expected that the PM will pull up a live view of the tool being discussed (i.e. during the Gantt discussion, the live Gantt chart should be pulled up and displayed, same for WBS). Judges will evaluate both the content and the team's ability to clearly communicate their approach and exemplified proficiency of each tool. Each section will have a 5 min presentation with a 5 min Q&A about that section along with a summary 5 min presentation and an extended Q&A after that. This will be scheduled to be 1 hour long.

The level of detail and rigor expected in each tool remains unchanged from previous years. Teams should not assume that the presentation format makes the deliverables easier; in fact, expectations for clarity, completeness, proficiency, and professionalism are higher.

Analyze and summarize the following for EACH deliverable in your winter workshop **presentation**:

- Show the deliverable live – i.e. pull up your real Gantt chart, Decision Making Model, etc.
 - Can use snips in a PowerPoint for the presentations but the live version must be up and active so it can be referenced for the judges during Q&A
- Compare what you did last year to what you have for this. What is the same? Why did you keep it? What is different? Why did you change it?
- Evaluate the tools you used. Are they the same as last year? Why / why not?
- What went well? What didn't go well?
- Based on last year's performance, what did you change in PM to address any & all shortcomings?

Format: 1 hour Time Slot: You will walk through each deliverable pausing between each for judges questions. You can use up to 5 minutes to present your deliverable before the judges stop you for questions. You do not need to use the full 5 minutes for each.

- Intro- up to 5 minutes
- Gantt Chart -up to 5 minutes
 - Judges questions – 5 minutes
- WBS- up to 5 minutes
 - Judges questions 5 minutes
-Continued for other deliverables.....
- Summary – up to 5 minutes
 - Final comments/questions from the judges

Limited to 3 people presenting, the project manager (or the interim) must be present as one of those representatives. However, the full team is encouraged to come sit in the audience to support their team while presenting. Members from other teams cannot be present.

Final Competition

Teams submit a summary that explains what changed from winter workshop to the final for each deliverable and an overall summary on how the team thinks they did this year/will do at the final competition. Additionally, the team is asked to reflect on how they did across all 5 years of series two. This should not be a full report, only a summary with a maximum of 5 pages (not including a cover page or table of contents). No appendices. This will be due prior to the final competition to allow time for the judges to review prior to the presentation.

At the final competition teams present their deliverables to judges and Q&A session. The format of this session will be 8 minutes for the team to present and the remaining time for the judges to ask questions. Please keep in mind that the judges have read your report and summary, so the material that is presented should be supplemental to the report.

Limited to 3 people presenting, the project manager (or the interim) must be present as one of those representatives. However, the full team is encouraged to come sit in the audience to support their team while presenting. Members from other teams cannot be present.

Note: Understanding that AutoDrive series II is ending, it is foreseeable that the PM of a team might have already left the team for various reasons. If this occurs, the team must pick a replacement and have that person be the PM for the final presentation. If this occurs, it should be discussed in detail during the presentation time.

PM.1.3.2 Scoring Rubric

The judges will evaluate the project management deliverables based on the program Leadership scoring rubric found at <http://www.AutoDrivechallenge.com>.

Deliverable	Description	Total Point Value
Gantt Chart	Winter Workshop - 5min max presentation, 5 min max questions. - The PM should pull up the team's Gantt chart and walk through it live. - Detailed description of Gantt Chart created for year 5 that includes: - All AutoDrive milestones (each deliverable, each challenge, check-in points and final competition) - Bar graphs between the milestones with team member names attached to them, target and actual dates? - All interdependencies included (all tasks linked) - Is the critical path defined (shown in red) - Did they describe their detailed critical path analysis? - Did they answer the judges' questions sufficiently?	10
WBS	Winter Workshop – 5 min max presentation, 5 min max questions. - The PM should pull up the team's WBS and walk through it live. - Detailed description of their WBS for each team member including work completed prior to Winter workshop and projected work after. - Are all Gantt chart items tied to WBS elements (people or things)? - How are people currently or projected to be over/underutilized? How did you adjust for it? Provide examples of how you used the WBS. - Names and hrs/wk assigned to deliverables. - Did they answer the judges' questions sufficiently?	10
Team Structure and Communication	Winter Workshop – 5 min max presentation, 5 min max questions. - Pull up the Org Chart and walk through it. Then cover team communication. - Did they cover the organizational chart? - Does it include every team member? - Side chart showing any relevant supporting individuals - Academic advisors, technical advisors, lab technicians, etc. - Team roles are clearly defined with description - Project manager is specifically defined - Did they cover their communication model? - Defined ways of communication and description of how each were used - How did you communicate with people outside of your team? How do you plan to do this in the future? - Give information on how your change control process was implemented - A full meeting schedule for how work was accomplished is included	10

	○ Garage, Lab, and test track schedule is included • Did they answer the questions sufficiently?	
Decision-Making Model (DMM)	Winter Workshop - 5 min max presentation, 5 min max questions • The PM should pull up DMM model and talk to it. ○ Is a model included? ○ Can members reference it for all decisions? Does it include sub teams/multiple decision makers? ○ Does the decision model match the organizational chart or is it different? Is there an explanation as to why/why not? ○ Includes examples of how the DMM was used? • Did they answer the judges' questions sufficiently?	10
Summary	Winter Workshop – 5 min max presentation, 10 min max questions • Did they correctly summarize the entire project? ○ General overview of year 5 performance using PM terms ○ How is the team doing? How does the team think they will perform in year 5 final competition? What challenges does the team foresee for the remainder of the year? ○ What is going well this year? What isn't going well this year? • Did they answer the judges' questions sufficiently?	10
Final Competition Report	Maximum of 5 pages, not counting cover page or table of contents. Brief overview for each of the 4 deliverables • Explain any changes made over the year, especially any changes made between winter workshop and final competition. Or, if no changes were made, explain how the tools worked well (why were changes not needed). • Give an overall summary of how well each of the PM tools worked throughout this year. Gantt (2 points) WBS (2 points) Team Structure and Communication (2 points) DMM (2 points) Final Summary (4 Points) • How does the team think they did this year? How do you think the team will perform at final competition? • Total 5-year reflection, how satisfied is the team with their entire AutoDrive Series 2 performance?	12
Final Competition Oral Presentation and Q&A	The PM must be present. 8 min presentation covering: • Year 5 • The entire 5 years of AutoDrive Series II. Be reflective of how the team did year-by-year compared to expectations. Hint, the communication plan has directed the teams to archive each year.	13

	Use PM language and tools to show what was expected vs what is/was reality, is there anything you would have done differently.	
Total	PM Total	75

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